



Time Series Analysis

Seminar 7: Cointegration and VECM



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Acronyms used in this material

- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ **ECM**: Error Correction Model
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GDP**: Gross Domestic Product
- ▣ **HUF**: Hungarian forint (ISO 4217 code)
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PLN**: Polish zloty (ISO 4217 code)
- ▣ **ROBOR**: Romanian Interbank Offer Rate
- ▣ **SE**: Standard Error
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VECM**: Vector Error Correction Model

Today's question and route

- **Question:** are two or more $I(1)$ series tied by a long-run equilibrium, how fast do they return to it, and who does the adjusting?
 - ▶ this seminar comes **before** Lecture 7: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: Engle–Granger statistics by hand; an ECM and its half-life; a bivariate VECM; Johansen statistics; from a VAR to a VECM
 - ▶ Part B: US interest rates; Romanian consumption and GDP; EUR/RON, EUR/HUF and EUR/PLN; ROBOR, the Romanian 10-year yield and Euribor
 - ▶ Part C: pairs trading on two Romanian banks, and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	an Engle–Granger statistic with two and with three variables	Solved, Proposed	A1
A3, A4	an ECM and a bivariate VECM: speed and half-life	Solved, Proposed	A3
A5, A6	Johansen statistics from eigenvalues; a VAR(2) written as a VECM	Solved, Proposed	A5
B1, B2	Engle–Granger and ECM: US yields; Romanian consumption and GDP	Solved, Proposed	B1
B3, B4	Johansen and VECM: three currencies; three interest rates	Solved, Proposed	B3
C1, C2	pairs trading out of sample; what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
US Treasury yields, 1 and 10 years	FRED (GS1, GS10)	monthly	1960–2026
Household consumption and GDP, Romania	Eurostat (namq_10_gdp)	quarterly, SA	1995–2026
EUR/RON, EUR/HUF, EUR/PLN	BNR reference rates	month-end	2005–2026
ROBOR 3M, Euribor 3M	Eurostat (irt_st_m)	monthly	2010–2026
Romanian 10-year yield	EODHD	daily, monthly mean	2010–2026
Banca Transilvania, BRD	EODHD	daily, adjusted close	2014–2026

- ▣ Levels as 100 ln (consumption, GDP, exchange rates, share prices); interest rates in % per year
- ▣ In the notebook: `adf_test`, `eg_test` (Engle–Granger and Phillips–Ouliaris), `ecm_fit`, `johansen`; no account or key is needed

Prerequisites for Today (1/5): cointegration

- $y_t \sim I(1)$: y_t has a unit root and $\Delta y_t = y_t - y_{t-1}$ is stationary (Chapter 3; ADF test, H_0 : unit root)
 - ▶ regressing one $I(1)$ series on another, unrelated one gives a **spurious regression**: large t , high R^2
- **Cointegration**: $I(1)$ series $y_{1t}, \dots, y_{nt}$ for which a combination $\beta^\top y_t$ is stationary
 - ▶ $y_t = (y_{1t}, \dots, y_{nt})^\top$: the vector of the n series; $\top$: transpose
 - ▶ β : the cointegrating vector, normalised with one coefficient equal to 1; $u_t = \beta^\top y_t - \mu$: the equilibrium error, μ : its mean
 - ▶ the series share common stochastic trends: n variables, r cointegrating vectors, $n - r$ common trends
- Example: the 1-year and the 10-year interest rate wander, but their spread stays within a band

Prerequisites for Today (2/5): the Engle–Granger test

- ▣ **Step 1:** OLS in levels, $y_t = a + bx_t + u_t$; residuals $\hat{u}_t = y_t - \hat{a} - \hat{b}x_t$
- ▣ **Step 2:** an ADF test on $\hat{u}_t$, without constant
 - ▶ $\Delta\hat{u}_t = \gamma\hat{u}_{t-1} + \sum_{j=1}^k \delta_j\Delta\hat{u}_{t-j} + e_t$, statistic $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$
 - ▶ δ_j : coefficients of the k lagged differences; e_t : white noise; SE: standard error
 - ▶ H_0 : no cointegration ($\gamma = 0$); reject if τ is below the critical value
- ▣ **Critical values** (MacKinnon, with a constant, large T): more negative than Dickey–Fuller, because OLS makes $\hat{u}_t$ look stationary
 - ▶ 5%: -2.86 (one series, Dickey–Fuller); -3.34 (2 variables); -3.74 (3 variables)
- ▣ **Phillips–Ouliaris** Z_t : the same H_0 and critical values; no lags, a long-run variance correction
 - ▶ in Python: `coint (statsmodels)`, `phillips_ouliaris (arch)`

Prerequisites for Today (3/5): error correction

- **ECM** (error correction model): the change of y reacts to last period's distance from equilibrium
 - ▶ $\Delta y_t = c + \gamma \hat{u}_{t-1} + \delta_0 \Delta x_t + \dots + \varepsilon_t$, with $\hat{u}_{t-1} = y_{t-1} - \hat{a} - \hat{b}x_{t-1}$
 - ▶ $\gamma < 0$: **speed of adjustment**, the share of last period's gap closed by y ; δ_0 : short-run effect; b : long-run effect
 - ▶ c : constant; "...": lags of Δy and Δx ; ε_t : white noise
- **Half-life**: a gap shrinks like $(1 + \gamma)^h$; half of it is gone after $h_{1/2} = \ln 0.5 / \ln(1 + \gamma)$ periods
 - ▶ example: $\gamma = -0.5$ gives $h_{1/2} = 1$; $\gamma = -0.1$ gives about 6.6 periods
- All the terms of an ECM are stationary: OLS and t -tests are valid

Prerequisites for Today (4/5): the VECM

▣ **VECM** (vector error correction model): a VAR(p) of [Chapter 6](#) rewritten in differences

- ▶ $\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + u_t$
- ▶ from $y_t = A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t$: $\Pi = A_1 + \dots + A_p - I$, $\Gamma_i = -(A_{i+1} + \dots + A_p)$
- ▶ A_i : the $n \times n$ VAR matrices; I : the identity matrix; u_t : white-noise shocks

▣ $r = \text{rank}(\Pi)$: the number of cointegrating vectors

- ▶ $r = 0$: no cointegration; $0 < r < n$: $\Pi = \alpha\beta^\top$, with α, β of size $n \times r$
- ▶ columns of β : the cointegrating vectors; α : the adjustment coefficients (loadings), how strongly each variable reacts to each equilibrium error
- ▶ $\alpha_i = 0$: y_i does not adjust (**weakly exogenous**)

Prerequisites for Today (5/5): the Johansen tests

- ▣ **Johansen:** maximum likelihood for the VECM gives eigenvalues $1 > \hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_n \geq 0$
 - ▶ a large $\hat{\lambda}_i$: a stationary combination of the levels; $\hat{\lambda}_i \approx 0$: a combination that is still $I(1)$
- ▣ Two statistics, with T the number of observations
 - ▶ trace: $-T \sum_{i>r} \ln(1 - \hat{\lambda}_i)$, H_0 : rank $\leq r$; maximum eigenvalue: $-T \ln(1 - \hat{\lambda}_{r+1})$, H_0 : rank $= r$
 - ▶ large statistics (eigenvalues far from 0): reject H_0
- ▣ **Sequential procedure:** test $r = 0, r \leq 1, \dots$; the rank is the first r not rejected
 - ▶ 5% critical values with a constant, $n = 3$: trace 29.80, 15.49, 3.84; max 21.13, 14.26, 3.84

A1: an Engle–Granger statistic [Solved]

Task

- ▣ Two $I(1)$ series, $T = 200$. Step 1 gives residuals $\hat{u}_t$; step 2 gives $\Delta\hat{u}_t = -0.118\hat{u}_{t-1}$, with $SE = 0.036$.
- ▣ 1. Compute τ .
- ▣ 2. Compare τ with the 5% and 10% Engle–Granger values for 2 variables, -3.37 and -3.07 , and decide.
- ▣ 3. Say what the Dickey–Fuller value -2.88 would have concluded.
- ▣ 4. Explain why the two critical values differ.
- ▣ Report: one number, two decisions and one sentence.

Solution

- ▣ 1. $\tau = -0.118/0.036 = -3.28$
- ▣ 2. $-3.28 > -3.37$: do not reject “no cointegration” at 5% ($p = 0.058$); $-3.28 < -3.07$: reject at 10%
- ▣ 3. $-3.28 < -2.88$: the Dickey–Fuller table would wrongly report cointegration at 5%
- ▣ 4. OLS chooses $\hat{b}$ to minimise the residual variance, so $\hat{u}_t$ looks more stationary than a true error: the null distribution shifts to the left.

A2: three variables [Proposed]

Solution discussed in class

Task

- ▣ Three $I(1)$ series, $T = 150$: y_t regressed on x_{1t} and x_{2t} . Step 2 gives $\hat{\gamma} = -0.162$, $SE = 0.041$. Model: A1.
- ▣ 1. Compute τ .
- ▣ 2. Decide at 5% and at 1% with the values for 3 variables, -3.80 and -4.39 .
- ▣ 3. Say which critical value a student would wrongly use if he forgot x_{2t} (2 variables: -3.38).
- ▣ 4. Say how many cointegrating vectors the Engle–Granger method can find with three variables.
- ▣ Report: one number, two decisions and two sentences.



A3: an error correction model [Solved]

Task

- ▣ Estimated ECM (log consumption c_t , log income y_t , in %): $\Delta c_t = 0.2 + 0.5 \Delta y_t - 0.25 (c_{t-1} - 0.9 y_{t-1})$.
- ▣ 1. Give the long-run and the short-run effect of income on consumption.
- ▣ 2. Interpret the coefficient -0.25 and compute the half-life.
- ▣ 3. Compute Δc_t if $c_{t-1} - 0.9 y_{t-1} = 2$ and $\Delta y_t = 1$.
- ▣ 4. Say what a coefficient of $+0.25$ would mean.
- ▣ Report: two elasticities, one half-life, one forecast and one sentence.

Solution

- ▣ 1. Long run: 0.9 (from the equilibrium $c = 0.9y + \text{const.}$); short run: 0.5 (same period)
- ▣ 2. A quarter of last period's gap is closed each period;
 $h_{1/2} = \ln 0.5 / \ln 0.75 = 2.41$ periods
- ▣ 3. $\Delta c_t = 0.2 + 0.5 \cdot 1 - 0.25 \cdot 2 = 0.2$: the error correction cancels the short-run effect
- ▣ 4. Consumption would move away from equilibrium after a gap: no error correction, no cointegration.

A4: a bivariate VECM [Proposed]

Solution discussed in class

Task

- ▣ $\Delta y_{1t} = -0.20 z_{t-1} + u_{1t}$, $\Delta y_{2t} = 0.05 z_{t-1} + u_{2t}$, with $z_t = y_{1t} - y_{2t}$. Model: A3.
- ▣ 1. Write α , β and $\Pi = \alpha\beta^\top$, and give the rank of Π .
- ▣ 2. Explain which variable does most of the adjusting.
- ▣ 3. Show that $z_t = (1 + \beta^\top \alpha) z_{t-1} + (u_{1t} - u_{2t})$ and compute its half-life.
- ▣ Report: one matrix, one rank, one half-life and one sentence.

A5: Johansen statistics from eigenvalues [Solved]

Task

- ▣ Three $I(1)$ series, $T = 200$, case with a constant; the Johansen eigenvalues are 0.120, 0.045 and 0.008.
- ▣ 1. Compute the trace statistics for $r = 0$, $r \leq 1$ and $r \leq 2$.
- ▣ 2. Compute the maximum-eigenvalue statistics.
- ▣ 3. Compare with the 5% values (trace 29.80, 15.49, 3.84; maximum 21.13, 14.26, 3.84) and choose the rank.
- ▣ 4. Give the number of common trends.
- ▣ Report: six statistics, one rank and one number of trends.

Solution

- ▣ 1. $-200[\ln 0.880 + \ln 0.955 + \ln 0.992] = 36.38$;
 $-200[\ln 0.955 + \ln 0.992] = 10.82$; $-200 \ln 0.992 = 1.61$
- ▣ 2. 25.57; 9.21; 1.61
- ▣ 3. $r = 0$ rejected by both ($36.38 > 29.80$, $25.57 > 21.13$);
 $r \leq 1$ not rejected: rank 1
- ▣ 4. $n - r = 3 - 1 = 2$ common stochastic trends.



A6: from a VAR(2) to a VECM [Proposed]

Solution discussed in class

Task

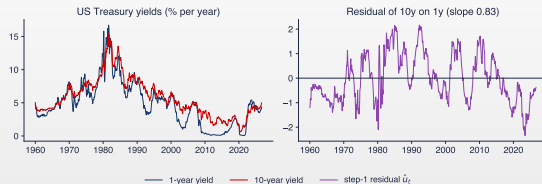
- ▣ $y_t = A_1 y_{t-1} + A_2 y_{t-2} + u_t$ with $A_1 = \begin{pmatrix} 0.7 & 0.2 \\ 0.1 & 0.7 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0.1 & 0 \\ 0 & 0.2 \end{pmatrix}$. Model: A5.
- ▣ 1. Compute $\Pi = A_1 + A_2 - I$ and $\Gamma_1 = -A_2$.
- ▣ 2. Show that Π has rank 1 and write it as $\alpha\beta^\top$ with $\beta = (1, -1)^\top$.
- ▣ 3. Interpret the signs of α .
- ▣ Report: two matrices, α and one sentence.



B1: the 1-year and the 10-year US yields [Solved]

- **Question:** are the 1-year and the 10-year Treasury yields cointegrated, how fast does their gap close, and which rate closes it?
- **Inputs:** FRED GS1 and GS10, monthly averages, % per year, January 1960–September 2026
- **Tasks**
 1. Run ADF (with a constant) on both yields and on the change of the 10-year yield.
 2. Run Engle–Granger (10-year on 1-year) and Phillips–Ouliaris, and compare with the 5% Engle–Granger value.
 3. Estimate the ECM of the 10-year yield (one lag of each difference) and the same equation for the 1-year yield.
 4. Interpretation: if the 10-year yield is 1 percentage point above its equilibrium, how long until half of the gap is closed, and by which rate?
- **Report:** three ADF p-values, two cointegration tests, two adjustment coefficients and one sentence
- **Notebook:** section B1

B1: solution [Solved]



- ▣ ADF $p = 0.26$ (1 year), 0.63 (10 years): unit roots; the change: $p < 0.001$; both yields $I(1)$
- ▣ Step 1: $\hat{i}_t^{(10)} = 1.76 + 0.827 i_t^{(1)}$; step 2: $\tau = -3.59$ (19 lags) < -3.34 : cointegration at 5% ($p = 0.025$); Phillips–Ouliaris $Z_t = -3.65$ ($p = 0.022$)
- ▣ ECM, 10 years: $\hat{\gamma} = -0.0295$ ($t = -4.81$), $\hat{\delta}_0 = 0.53$; 1 year: $\hat{\gamma} = 0.0172$ ($t = 1.25$), not significant
- ▣ Interpretation: half-life $\ln 0.5 / \ln(1 - 0.0295) = 23.1$ months, about two years; the 10-year yield closes the gap, the 1-year yield (set by monetary policy) does not react

B2: consumption and GDP in Romania [Proposed]

- **Question:** is Romanian household consumption cointegrated with GDP, as the “great ratios” suggest?
- **Inputs:** Eurostat, real household consumption and real GDP, chain-linked volumes, seasonally adjusted, 1995–2026; c_t , y_t as 100 ln; model: B1
- **Tasks**
 1. Run ADF (constant and trend) on c_t and y_t , and ADF (constant) on their changes.
 2. Run Engle–Granger (c_t on y_t) and Phillips–Ouliaris, and report the slope.
 3. Plot the ratio of consumption to GDP and run ADF on $c_t - y_t$.
 4. Interpretation: does a stable consumption share make sense for Romania over 1995–2026?
- **Report:** four ADF tests, two cointegration tests, one chart and two sentences
- **Notebook:** section B2

B3: three Central European currencies [Solved]

- ▣ **Question:** do EUR/RON, EUR/HUF and EUR/PLN share a long-run equilibrium?
- ▣ **Inputs:** BNR reference rates (cross rates for HUF and PLN), month-end, 100 ln, since July 2005
- ▣ **Tasks**
 1. Run ADF on each rate.
 2. Run the Johansen trace and maximum-eigenvalue tests with a constant and lags by BIC.
 3. Compare with the three pairwise Engle–Granger tests and with the correlations of levels and of changes.
 4. Interpretation: should the three rates be modelled with a VECM?
- ▣ **Report:** three ADF p-values, a table of six Johansen statistics, a rank and one sentence
- ▣ **Notebook:** section B3

B3: solution [Solved]



	$r = 0$	$r \leq 1$	$r \leq 2$
trace (5% value)	24.21 (29.80)	7.94 (15.49)	2.72 (3.84)
maximum eigenvalue (5% value)	16.27 (21.13)	5.21 (14.26)	2.72 (3.84)

- ADF $p = 0.55; 0.75; 0.08$: three $I(1)$ rates; Johansen ($T = 255, k = 1$): no statistic exceeds its critical value: rank 0
- Engle–Granger $p = 0.29$ (RON, HUF), 0.08 (RON, PLN), 0.20 (HUF, PLN); correlation 0.89 in levels but only 0.31 in monthly changes
- Interpretation: no; with rank 0 there is no β to estimate; model the changes with a VAR in differences ([Chapter 6](#))

B4: ROBOR, the Romanian 10-year yield and Euribor [Proposed]

- **Question:** how many long-run relations tie Romanian interest rates to each other and to Euribor, and which rate adjusts?
- **Inputs:** ROBOR 3M and Euribor 3M (Eurostat), Romanian 10-year yield (monthly mean of daily data), January 2010–August 2026; model: B3
- **Tasks**
 1. Run the Johansen tests (constant, BIC lags) and choose the rank.
 2. Estimate the VECM with that rank and the constant restricted to the cointegrating relation; normalise β on ROBOR.
 3. Report $\hat{\alpha}$ with t -statistics and run ADF on the equilibrium error.
 4. Interpretation: which rates are weakly exogenous, and what does this say about Romanian monetary conditions?
- **Report:** a table of Johansen statistics, $\hat{\beta}$, $\hat{\alpha}$ with t , one ADF test and two sentences
- **Notebook:** section B4

C1: pairs trading on two banks [Proposed]

- **Question:** would a trader who found Banca Transilvania and BRD cointegrated at the end of 2021 have earned money afterwards?
- **Inputs:** adjusted daily prices, formation 2014–2021, trading January 2022–September 2026; models: B1, B3
- **Tasks**
 1. On the formation period, run Engle–Granger and Phillips–Ouliaris (TLV on BRD) and keep $\hat{b}$, the mean and the standard deviation of the spread.
 2. On the trading period, apply the rule: open at $|z| > 2$, close at $z = 0$ (z : the spread minus its formation mean, divided by its formation standard deviation); compute the annual return and the Sharpe ratio (mean annual return divided by its annual standard deviation), gross and net of 0.20% per unit traded.
 3. Repeat the Engle–Granger test on the trading period alone.
 4. Interpretation: is the result evidence that pairs trading works on the BVB?
- **Report:** two tests, the number of trades, two returns, two Sharpe ratios and a plan for a project
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant about cointegration. The answer:
- (a) The Engle-Granger statistic is -3.0 with two variables; since $-3.0 < -2.87$, the 5% Dickey-Fuller value, the series are cointegrated.
- (b) The ECM coefficient of the lagged equilibrium error is $+0.15$, so 15% of the gap is corrected each period.
- (c) Johansen rejects $r = 0$ and $r \leq 1$ but not $r \leq 2$ for three yields, so the yields share two common trends.
- (d) EUR/RON and EUR/HUF are both $I(1)$ and $R^2 = 0.79$ in levels, so they are cointegrated.
- (e) If two series are cointegrated, at least one of them Granger-causes the other.
- (f) A pairs-trading rule with a Sharpe ratio of 0.44 on 2014-2026 data will keep this Sharpe ratio in the future.
- **Tasks**
 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- **Report:** a list of six verdicts with one line of justification each

Key takeaways

- ▣ Test that every series is $I(1)$ first; then test the equilibrium error, never the correlation of levels
- ▣ Residual-based tests need Engle–Granger critical values, which depend on the number of variables
- ▣ The ECM gives the speed of adjustment; the half-life $\ln 0.5 / \ln(1 + \gamma)$ is in the units of the data
- ▣ Johansen gives the rank; $n - r$ is the number of common trends; α shows who adjusts
- ▣ An AI answer is a draft: check the critical values, the signs and the number of trends

After the seminar

- Lecture 7 develops each topic of today
 - ▶ common trends, the Engle–Granger distribution, ECM, the VECM and the Granger representation theorem, the Johansen tests, forecasting and pairs trading
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: pairs trading on the BVB, out of sample and after costs
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 9; [Hamilton \(1994\)](#), Ch. 19–20; [Engle and Granger \(1987\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

- ▣ Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276. doi:[10.2307/1913236](https://doi.org/10.2307/1913236)
- ▣ Gatev, E., Goetzmann, W. N., & Rouwenhorst, K. G. (2006). Pairs trading: Performance of a relative-value arbitrage rule. *Review of Financial Studies*, 19(3), 797–827. doi:[10.1093/rfs/hhj020](https://doi.org/10.1093/rfs/hhj020)
- ▣ Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:[10.2307/j.ctv14jx6sm](https://doi.org/10.2307/j.ctv14jx6sm)
- ▣ Huang, C., & Petukhina, A. (2022). *Applied Time Series Analysis and Forecasting with Python*. Springer. doi:[10.1007/978-3-031-13584-2](https://doi.org/10.1007/978-3-031-13584-2)
- ▣ Johansen, S. (1988). Statistical analysis of cointegration vectors. *Journal of Economic Dynamics and Control*, 12(2–3), 231–254. doi:[10.1016/0165-1889\(88\)90041-3](https://doi.org/10.1016/0165-1889(88)90041-3)
- ▣ Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1580. doi:[10.2307/2938278](https://doi.org/10.2307/2938278)
- ▣ MacKinnon, J. G. (2010). Critical values for cointegration tests. Queen's Economics Department Working Paper No. 1227, Queen's University. ideas.repec.org/p/qed/wpaper/1227
- ▣ Phillips, P. C. B., & Ouliaris, S. (1990). Asymptotic properties of residual based tests for cointegration. *Econometrica*, 58(1), 165–193. doi:[10.2307/2938339](https://doi.org/10.2307/2938339)